Swati Aggrawal

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RESEARCH PROFILE

M.Sc. Computer Science graduate (AI/ML specialization) from Sapienza Università di Roma with a peer-reviewed IEEE publication and a master's thesis on continual learning for Vision Transformers, achieving a ~97% relative reduction in catastrophic forgetting versus a CNN baseline. Combines research rigor in machine learning and algorithmic evaluation with hands-on engineering — building, benchmarking, and deploying ML systems in Python and C++ on AWS. Seeking a Research Sciences internship to contribute to applied research in machine intelligence.

SKILLS

- **Research & ML:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, Algorithm Design & Analysis, Experimental Evaluation, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
- **Programming:** Python, C++, Java, SQL, Bash
- **Systems & Cloud:** AWS (ECS, Fargate, CloudWatch), Docker, Linux, FastAPI, REST APIs, Git
- **Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), AI Agents, Vector Databases (Pinecone), Gemini API, Prompt Engineering

RESEARCH EXPERIENCE

Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis)

Nov 2025 – May 2026

Advisor: Prof. Marco Raoul Marini | https://github.com/swatiaggrawal/Thesis-Vision-Transformers

- Researched continual learning for computer vision, designing and running controlled experiments to isolate the effect of adapter-based fine-tuning and Elastic Weight Consolidation (EWC) on catastrophic forgetting.
- Built PyTorch training and evaluation pipelines benchmarking accuracy, forgetting, and computational efficiency across sequential classification tasks on Split CIFAR-100 (5 sequential tasks).
- **Achieved 90.59% average accuracy with only 2.01% average forgetting**, a ~97% relative reduction in catastrophic forgetting versus a naive ViT fine-tuning baseline (41.91% accuracy) and a CNN/ResNet-18 baseline (24.22% accuracy).
- Benchmarked against ResNet-18 and naive ViT fine-tuning baselines, validating results via cross-seed evaluation.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.

Technologies: Python, PyTorch, Vision Transformers, Elastic Weight Consolidation, Computer Vision

PUBLICATION

Machine Learning for Air Pollution — IEEE ViTECoN

2023

doi.org/10.1109/ViTECoN58111.2023.10157028

- Authored a peer-reviewed IEEE survey paper reviewing six ML models (ANN, GA-ANN, Random Forest, Decision Tree, SVM, LS-SVM) used for air quality prediction, synthesizing findings from 35+ cited studies
- Reviewed air quality evaluation standards, including the six NAAQS criteria pollutants and their primary/secondary thresholds
- Analyzed prediction-based (classification/regression) and knowledge-discovery (clustering) learning paradigms applied in air pollution epidemiology
- Synthesized recent advances across ~12 published studies (2018–2022), covering methods like LightGBM, LSTM with genetic algorithms, and hybrid interpretable models
- Identified future research directions, including real-time IoT-integrated air quality monitoring and the shift toward deep learning approaches

APPLIED RESEARCH & ENGINEERING PROJECTS

LossyTextCompressor — Semantic Embedding Quantization

github.com/swatiaggrawal/LossyTextCompressor

- Studied the trade-off between compression rate and semantic preservation by generating and quantizing transformer text embeddings, then reconstructing text from compressed vectors.
- Evaluated reconstruction quality across compression levels using ROUGE-L and BLEU, quantifying the accuracy/efficiency frontier for embedding-based text compression.

Technologies: PyTorch, GTR-T5, Vec2Text, Quantization, ROUGE-L, BLEU

Knowledge Copilot — Grounded RAG Chatbot

github.com/swatiaggrawal/RAG-Chatbot | Live demo: rag-chatbot-new.onrender.com

- Built an end-to-end retrieval-augmented generation (RAG) chatbot that answers questions over FastAPI's official documentation, with every response grounded in retrieved source text and returned with per-answer citations to prevent hallucination.
- Architected a modular four-stage pipeline (ingestion, embedding, retrieval, generation), each independently runnable and testable via CLI, isolating failures to a specific stage.
- Implemented recursive paragraph/sentence-aware chunking with Unicode normalization to fix PDF ligature artifacts, improving embedding quality over naive fixed-size character splitting.
- Integrated Gemini API embeddings (task-type-aware asymmetric embeddings for queries vs. documents) with Pinecone as a managed vector store, removing a local sentence-transformers/torch dependency to fit the deployed backend within a 512MB free-tier RAM limit.
- Built a FastAPI backend exposing a /chat endpoint and a build-free React (CDN) frontend displaying source citations per answer; implemented self-indexing startup logic that auto-runs the ingest→embed→store pipeline on a fresh deploy.

Technologies: Python, FastAPI, React, Gemini API, Pinecone, RAG, LLMs, Vector Databases

DocDelta — AI-Powered Document Intelligence & Comparison Platform

github.com/swatiaggrawal/DocDelta

- Developed an end-to-end semantic document comparison API using FastAPI and Sentence Transformers (all-MiniLM-L6-v2), detecting and classifying sentence-level changes (added/removed/modified, numerical vs. wording) beyond traditional character-level diffs.
- Designed scalable embedding and cosine-similarity retrieval pipelines, achieving ~840ms warm-request latency on CPU-only inference.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring for latency and error tracking.
- Built REST APIs with async request handling and validation, optimizing container size (CPU-only PyTorch builds) and eliminating cold-start overhead via model preloading.

Technologies: Python, FastAPI, Sentence Transformers, PyTorch, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

github.com/swatiaggrawal/Metal-Defect-Detection

- Built a custom CNN (TensorFlow/Keras) plus SVM and Random Forest classifiers (HOG features) for automated industrial metal surface defect detection across 6 defect classes on the NEU dataset.
- Built image preprocessing, augmentation, training, and evaluation pipelines, tuning classical models via GridSearchCV.
- **Achieved 97.22% test accuracy** with the CNN versus 86.11% for both SVM and Random Forest — an ~11-point improvement demonstrating the value of learned spatial features over hand-crafted descriptors.
- Evaluated models using accuracy, confusion matrices, precision, recall, and F1-score.

Technologies: Python, TensorFlow, Keras, OpenCV, Scikit-Learn

EDUCATION

Sapienza Università di Roma — M.Sc. Computer Science 2026

- Relevant Coursework: Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning

Sharda University — B.Tech, CSE with AI and ML 2023

- Relevant Coursework: Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- Awarded merit-based scholarship (Top 5–10% of cohort)